

When AI Thinks With Us

An Exploratory Probe into AI-Assisted Learning and Cognitive Delegation

T Fatimah | September 2026 | Exploratory pilot study, n = 5

Abstract

Generative AI can now perform work that people once did themselves, including explaining, summarising, problem solving and giving feedback. This raises a design question: when AI is available for part of a cognitive task, how do people decide what to keep and what to hand over? This report describes a small exploratory probe with five participants aged 19–25. Each answered a short scenario about a university student deciding how to use AI for examination revision, read one standardised AI recommendation, answered the same question again without the recommendation in view, and rated whether it had changed their understanding. The analysis is a descriptive reading of what participants wrote and does not test any hypothesis. Participants generally described AI as a support for clarification, difficult material and practice, and several set conditions around its use, including an independent attempt first. Three of five reported that the AI response changed their understanding somewhat, one very little and one not at all. These are self-reports. Because the recommendation itself endorsed an independent-first sequence, similarity between some later answers and that advice could reflect reconsideration, priming, conformity, prior belief or a mixture, and the data cannot separate them. The probe is offered as a first step towards a controlled study of how responsibility for cognitive work is allocated between people and AI.

1. Motivation and Research Context

I came to this question through practice rather than through the literature. Over more than fifteen years in enterprise technology, my work moved from systems and digital transformation to workflow and process improvement, Agile transformation and product management, and more recently to AI-enabled products, workflows and agents, including a small agent I built to explore how tasks are handed over in knowledge work. As the systems I worked with began to carry out parts of the thinking, one question interested me more and more, and delivery metrics do not answer it. What happens to the person in the workflow, and to their learning, judgement and sense of ownership, when AI stops being a tool that is consulted and becomes a contributor to cognitive work?

Research on cognitive offloading is a natural starting point. Risko and Gilbert (2016) define it as the use of physical action to alter the information-processing demands of a task so as to reduce cognitive demand, and review what triggers offloading and what follows from it. Two recent survey studies bring related ideas to generative AI in learning. Hashmi et al. (2026), surveying 861 students, report a positive association between generative AI use and perceived self-regulated learning, with technological self-efficacy and cognitive offloading as mediators; students in the same study also raised over-reliance and metacognitive laziness as concerns. Zhu et al. (2026) separate dependent offloading, in which core thinking is delegated to AI, from autonomous offloading, in which AI scaffolds thinking while the user keeps cognitive agency, and report different associations with perceived cognitive outcomes in a three-wave survey of 589 students and early-career workers. Both are survey studies based on self-report, and Zhu et al. describe their own evidence as correlational.

The present pilot does not ask whether AI helps or harms learning. It asks how a small group of people describe the line between their own effort and AI assistance, and it is meant to help turn a broad practical concern into a question that can be studied. The wider question behind this work is how AI-augmented work systems can be designed to improve human performance and collaboration without unintentionally increasing cognitive load, reducing learning, or weakening human judgement and agency. The pilot does not answer that question.

2. Research Questions

RQ1. How do participants describe appropriate uses of AI when a learner is deciding how much of their studying to do independently?

RQ2. After exposure to a standardised AI recommendation, what changes, if any, appear in participants' descriptions of appropriate AI use?

RQ3. What early patterns suggest how people may decide which parts of cognitive work to retain and which to delegate to AI?

3. Method

3.1 Participants

Five participants (P1–P5), aged 19–25, took part as a convenience sample. Their backgrounds spanned science, technology, infrastructure, commerce and business or entrepreneurial activity. Participants are identified only by anonymous codes, and no names were retained.

3.2 Scenario and Procedure

All participants received the same scenario, “The Overloaded Student” (Appendix A). Alex, a second-year university student, is preparing for examinations in four subjects, two weeks away, and is struggling with readings and assignments. Alex is considering using AI to summarise difficult readings, explain concepts and generate practice questions. Participants first answered Q1: “Based only on the scenario, what do you think are the THREE most important things Alex should consider when deciding how to use AI for studying?” They were then shown the same standardised AI response:

“AI can help Alex summarize difficult material, explain concepts and generate practice questions. However, Alex should continue attempting to understand and recall material independently because actively retrieving information and explaining concepts can support learning. A balanced approach could involve attempting a problem independently first, then using AI for clarification, feedback or additional practice.”

The response was then hidden, and participants answered the question again without looking back (Q2). Q3 asked whether the AI response had changed their understanding of how Alex should study, with four options: yes, substantially; yes, somewhat; very little; not at all. An optional final question asked which parts of their own studying they would do themselves and which they would delegate to AI, and why. Answers were given in writing.

3.3 Analytic Approach

The analysis is a descriptive thematic reading of the written answers, carried out by me alone. I read the responses, noted recurring ideas and grouped them into tentative themes. There was no second coder, no coding software and no formal reliability check. The aim was to describe how participants talked about the boundary between independent work and AI assistance, not to test hypotheses. Below, I keep three levels apart: what participants wrote (observed), my reading of patterns across them (interpreted), and what such patterns might suggest for future testing (hypothesised).

Participants were asked for three points in Q1 and Q2. P1 gave four in Q1 and two in Q2, and P4 gave more than three in Q2; all substantive responses were retained. One phrase in P4's Q2, “less emphasis on accuracy”, is ambiguous, and I have neither interpreted it nor drawn on it for any theme.

4. Findings

4.1 Self-Reported Change

Observed. Three of five participants selected “Yes, somewhat” (P3, P4, P5), one selected “Very little” (P2) and one “Not at all” (P1). No participant selected “Yes, substantially”.

Table 1. Self-reported change (Q3) and post-exposure answers.

Participant	Q3 response	Post-exposure answers (Q2 and optional question), summarised
P1	D, Not at all	Build basics first, then add own critical thinking with AI as support; test own understanding rather than accept generic advice. Would build basics with AI, then practise active recall in own style.

Participant	Q3 response	Post-exposure answers (Q2 and optional question), summarised
P2	C, Very little	Summarise concepts, generate practice questions, ask for steps and not only final answers. Would use AI to summarise difficult concepts, to understand them before studying in detail.
P3	B, Yes, somewhat	Try to solve independently first, understand concepts through AI, then practise. Easy parts done alone; difficult parts delegated to AI “because I’m not able to understand them”.
P4	B, Yes, somewhat	Summarise and explain difficult concepts; study independently to understand and recall; try to learn alone first. Would read notes, use active recall and solve problems, then take AI support, with AI-made practice questions for difficult topics.
P5	B, Yes, somewhat	A balanced approach: independent learning, with AI help if the material is tough; summarise notes, then ask for practice questions. Would use own notes and active recall first, then AI for practice and correction.

Interpreted. Q3 records how participants perceived their own change. It cannot show that their views changed, or that they did not, and participants may have judged their own change inaccurately. With $n = 5$ these counts are not estimates of anything beyond this group.

4.2 Recurring Themes

This section reports what participants wrote (observed). Interpretation follows in Section 4.3.

AI as support. P1 wrote that AI should support learning and not replace notes or a teacher, and P4 that AI should support learning. P5 described AI as a “learning partner” and a friend who would help correct and advise. P2 and P3 described specific uses (summarising, explaining, practice questions) without stating a view on replacement.

Independent attempt. In Q1, no participant described attempting the material before turning to AI as an explicit sequence. After the AI response, P3, P4 and P5 did so, in Q2 or in the optional question: P3 (“try to solve independently”), P4 (study independently to understand and recall; learn “by himself first”) and P5 (independent learning, with AI if it is “tough”). P1 wrote that basics should come first and that own critical thinking should follow, but said in the optional question that basics would be built with AI. P2 did not describe an independent step.

Difficulty. P3 stated a rule: easy parts alone, difficult parts delegated to AI, “because I’m not able to understand them”. P4 (“especially difficult topics”) and P5 (“if it is tough”) described turning to AI when they could not manage alone or when material was hard. P2 described AI as a way into difficult concepts before detailed study, which is a use for hard material but is not framed as a condition. The scenario and the AI response both mention difficult material, so this vocabulary was available to every participant.

Practice and recall. All five mentioned practice or recall in some form: active recall (P1, P4, P5), practice questions (P2, P4, P5), mock tests (P4, P5) and “practice learning” (P3). Practice questions appear in both the scenario and the AI response.

Verification and dependence. In Q1, P4 wrote that AI-generated summaries and practice questions can be wrong and should be checked against lecture notes; this did not reappear in P4’s Q2 in comparable terms. In Q2, P1 wrote that a learner should test their own understanding and not take generic advice from AI, a related point framed differently. Concern about dependence was explicit in P1 (“do not become dependent”) and P5 (“less dependent on AI itself”). Verification and dependence were raised by a minority of participants, and not by the same ones.

4.3 Emerging Interpretation

Interpreted. P3, P4 and P5 each expressed an attempt-first, AI-second sequence after the AI response, and each also described some conditional use of AI. These are the three who rated the change as “somewhat”. P1 and P2, who rated it “not at all” and “very little”, did not add an attempt-first step. P1’s answers stressed critical thinking and dependence from the start, and P2’s treated AI as a way into difficult material both before and after. One reading is that people take up advice more readily when it sits comfortably with how they already think about learning. Five participants cannot support that reading, and I do not treat it as a finding.

Hypothesised. Two propositions could be tested. First, that the perceived difficulty of a task predicts willingness to delegate it to AI. Second, that people's existing views on how learning works moderate how far they revise their plans after hearing AI advice. Risko and Gilbert (2016) note that offloading is shaped by the internal demands of a task and by metacognitive judgements of one's own ability, which can be mistaken, and this is one reason to expect perceived difficulty, rather than objective difficulty, to matter. Neither proposition has been tested here. Difficulty was neither measured nor manipulated, and stated preferences were not compared with behaviour.

5. Discussion

The most consistent observation is that participants placed conditions around AI use, and neither embraced nor rejected it. AI was welcomed for explanation, summarising, practice and feedback, and in four of five responses it was paired with independent effort, recall or checking. This is what participants said they would do in a hypothetical, or about a hypothetical student; it is not a record of what they did.

Cognitive offloading is a useful lens for the wider problem, but the pilot does not demonstrate it. Nothing was measured about cognitive effort, memory load or learning. I therefore use "AI-supported cognitive delegation" only as a descriptive phrase for participants' stated choices about which parts of a learning task they would keep and which they would ask AI to support. The distinction drawn by Zhu et al. (2026) between delegating core thinking and using AI as a scaffold while retaining agency resembles, at a descriptive level, what some participants said. The resemblance is loose, and the pilot cannot say whether their stated choices would be borne out in practice.

The standardised AI response recommended an independent-first, AI-supported approach, which creates the main methodological problem. Where later answers resemble it, several explanations remain open: genuine reconsideration; priming by wording seen moments earlier; memory of the response, even though it was hidden; conformity with an apparent authority or with what participants thought was expected; beliefs already held and now easier to state; or some combination. P5's Q2 begins with "balanced approach", close to the AI text. That proximity is an observation, and it is not evidence either of influence or of independent agreement. Nothing in the design allows me to say which explanation applies.

The pilot therefore raises a question and does not answer one. When AI assistance is sequenced or bounded differently, does the person keep more or less of the cognitive work, and does that matter for learning and judgement?

6. Limitations

The following limitations bound every statement above.

- The sample is five people, recruited by convenience and aged 19–25. It does not represent learners or workers more widely, and the study is exploratory and qualitative.
- The scenario concerned studying and not workplace collaboration. Any relevance to professional work is a proposition for research and not a finding.
- There was no control group. Without one, the effect of the AI response cannot be separated from the effect of answering the same question twice, or from wording in the scenario.
- No objective learning measure, delayed recall, transfer test, cognitive-load measure or observation of actual AI use was taken. Answers describe intentions about a hypothetical, and stated preferences may not match behaviour.
- Q3 measures perceived change only.
- Coding and interpretation were done by one researcher, and no inter-rater reliability was calculated.
- The standardised AI response may have primed later answers. Similarity afterwards could reflect reconsideration, priming, memory of the response, conformity, pre-existing beliefs or a mixture, and I do not claim to know which.
- Participants gave differing numbers of points (P1 four in Q1 and two in Q2; P4 more than three in Q2). All substantive responses were retained. One ambiguous phrase from P4 was not interpreted.

7. Proposed Next-Stage Study

Before any experiment, I would extend the qualitative work with interviews or think-aloud sessions across a more varied sample, coded independently by two people. The controlled study that the pilot points towards would then compare how the timing and role of AI assistance relate to what people retain of a task. Four conditions are possible:

- **Human-first:** the participant attempts the task, then receives AI assistance or feedback.
- **AI-first:** AI provides an explanation or solution, then the participant attempts the task.
- **Human–AI collaborative:** iterative interaction between participant and AI.
- **Human-only control:** no AI assistance.

Possible measures are task performance, immediate and delayed recall, transfer to a novel problem, perceived cognitive load, confidence and calibration, the amount and timing of AI assistance, willingness to delegate, and behavioural interaction logs. Perceived task difficulty and participants' prior views on learning, the two ideas raised in Section 4.3, could be examined as moderators. I would not assume the direction of any difference, since AI-first sequences may suit some learners or tasks. Sample size would be set by a power analysis, and hypotheses would be specified in advance. NeuroIS methods, including EEG, could later be considered where they answer a question that behavioural and self-report measures cannot. They are not needed to interpret this pilot, and I would seek training before proposing them.

8. Relevance to Human-Centered AI-Augmented Work

Although the probe uses a learning scenario, the design problem it points to extends to knowledge-intensive work, where AI can also explain, generate, summarise and recommend. The question is where the boundary between person and system should sit, when AI should intervene, and how a system can support performance without unintentionally weakening learning, judgement or agency.

This overlaps with the agenda of the Human-Centered Information Systems Engineering (HISE) group at the University of Zurich, which describes its aim as designing information systems that support, extend and responsibly orchestrate human collaboration and do not replace human intelligence (HISE, n.d.). Prof. Dr. Eva Bittner describes her research as studying and designing collaborative work practices and socio-technical systems for knowledge-intensive work, with attention to the social, cognitive and ethical implications of human–AI collaboration and to its unintended side effects (Bittner, n.d.). The current PhD call lists, among its topic areas, socio-cognitive mechanisms in human–AI collaboration and designing towards balanced engagement, including cognitive load and learning versus deskilling, as well as agentic information systems for human-centred work (University of Zurich, n.d.). The pilot touches the cognitive and learning side of that agenda at a very small scale.

It also leaves a good deal open. The pilot concerns individual students and not groups or teams, involves no collaboration task, measures no behaviour and evaluates no designed artefact. A doctoral project on AI-augmented work would need to move from stated preferences to observed behaviour in work-like settings, and from description to the design and evaluation of collaboration processes and tools. The call describes a project with room for the candidate to shape their own topic. I would expect to develop the question with the group and not to bring it fully formed.

9. Conclusion

This study began with a practical question: when AI can do part of a cognitive task, what do people choose to keep for themselves? Five participants described AI mainly as a source of clarification, explanation, practice and feedback, and most set conditions around its use. Three said the standardised response changed their understanding somewhat, but the design cannot show whether that reflected reconsideration, priming, conformity or earlier belief. What the pilot offers is a sharper question and an account of its own weaknesses: how the order and conditions of AI assistance affect what people retain of the work, and whether the answer depends on perceived difficulty and on a person's existing view of learning. Answering it will need controlled comparison, outcome measures beyond self-report, and methodological training that I hope to acquire through doctoral research.

References

- Bittner, E. (n.d.). *Research vision and short biography*. Human-Centered Information Systems Engineering, Department of Informatics, University of Zurich. <https://www.ifi.uzh.ch/en/hise/people/bittner.html> (accessed 24 September 2026).
- Hashmi, Z. F., Iqbal, J., Asghar, M. Z., & Abid, M. N. (2026). Generative AI tool usage and perceived self-regulated learning: The mediating roles of technological self-efficacy and cognitive offloading in blended learning. *Humanities and Social Sciences Communications*, 13, Article 1618. <https://doi.org/10.1057/s41599-026-07924-3>
- HISE. (n.d.). *Human-centered Information Systems Engineering*. Department of Informatics, University of Zurich. <https://www.ifi.uzh.ch/en/hise.html> (accessed 24 September 2026).
- Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676–688. <https://doi.org/10.1016/j.tics.2016.07.002>
- University of Zurich, Department of Informatics. (n.d.). *Open PhD position: Human-centered design of AI-augmented collaboration and work*. Human-Centered Information Systems Engineering. <https://www.ifi.uzh.ch/en/hise/openpositions/Open-PhD-Position--Human-Centered-Design-of-AI-Augmented-Collaboration-and-Work.html> (accessed 24 September 2026).
- Zhu, Q., Li, X., Dong, Y., Chang, P., & Fan, M. (2026). Not all cognitive offloading is equal: Distinguishing dependent and autonomous offloading to generative AI. *Frontiers in Psychology*, 17, 1878629. <https://doi.org/10.3389/fpsyg.2026.1878629>

Appendix A. Study Materials

Scenario: The Overloaded Student

Alex is a second-year university student preparing for final examinations in four subjects. With two weeks remaining, Alex is struggling to keep up with readings and assignments and often studies late at night while moving between notes, online resources and social media. Alex decides to create a weekly study plan, divide material into smaller sections, practise with questions and take regular breaks. Alex is also considering using AI to summarise difficult readings, explain concepts and generate practice questions, but is unsure how much of the studying should be done independently and how much should be delegated to AI.

AI response shown to all participants

AI can help Alex summarize difficult material, explain concepts and generate practice questions. However, Alex should continue attempting to understand and recall material independently because actively retrieving information and explaining concepts can support learning. A balanced approach could involve attempting a problem independently first, then using AI for clarification, feedback or additional practice.

Questions

Q1. Based only on the scenario, what do you think are the THREE most important things Alex should consider when deciding how to use AI for studying?

Q2. Without looking back at the AI response, what are the THREE most important things you now think Alex should consider?

Q3. Did the AI response change your understanding of how Alex should study? A. Yes, substantially. B. Yes, somewhat. C. Very little. D. Not at all. Briefly explain why.

Optional. If you were Alex, which parts of your studying would you prefer to do yourself, and which would you delegate to AI? Why?